DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

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Lecture 10: The Capital Asset Pricing Model

From Two-Fund Separation to Market Equilibrium · The CAPM Equation

August 12, 2026

Today's Agenda

1. From separation to equilibrium: the market portfolio

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4. CML vs. SML

Learning Objectives

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- State the CAPM equation and its key assumptions.
- Explain why only systematic risk is priced.
- Distinguish the CML from the SML.

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One Restaurant, One Town

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Every investor holds the risk-free asset and the same tangency portfolio T. Add up everyone's holdings of risky assets – that sum must equal the *entire* market of risky assets. There's nowhere else for those shares to be held. This forces T to **be** the market portfolio.

The Market Portfolio

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The Central CAPM Insight

Since every investor holds the same tangency portfolio, and markets must clear: $T = M$.

The Assumptions Behind This Leap

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Not literally true of any real market. The value is a simple, testable prediction – not realism.

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No Free Lunch for Diversifiable Risk

If a risk that's costlessly removed by diversifying still earned extra return, anyone could grab that reward for free. Competed away instantly. Risk you *can't* diversify away is a cost everyone bears together – only that risk earns a lasting reward.

Only Systematic Risk Is Priced

Key Takeaway

An asset's risk premium depends on its **covariance with the market**, $\mathrm{Cov}(R_i, R_m)$ – *not* its own total variance σ_i^2.

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Two assets, same σ_i, can have very different expected returns.

Beta and the CAPM Equation

Definition: Beta

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$$\mathbb{E}[R_i] = r_f + \beta_i\big(\mathbb{E}[R_m] - r_f\big)$$

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	β=1.3	β=0.6	β=1.0
$\mathbb{E}[R_i]$	13.125%	8.75%	11.25%

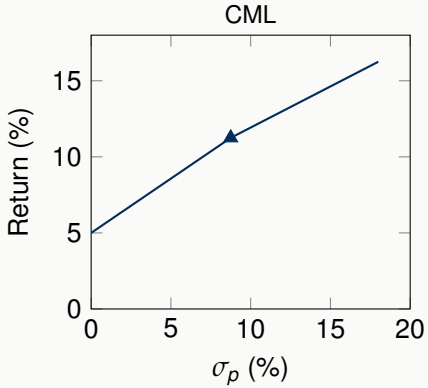
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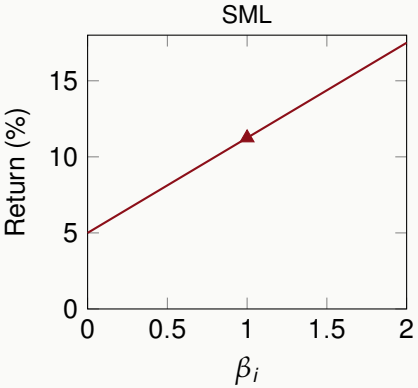
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$\beta = 1$ *earns exactly* $\mathbb{E}[R_m]$ *– moves exactly like the market.*

Two Lines, Two Different Questions





CML vs. SML

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A single stock's σ_i has unsystematic risk mixed in – SML strips that out with β_i.

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- Only **systematic risk** is priced – unsystematic risk earns no reward.
- **CAPM**: $\mathbb{E}[R_i] = r_f + \beta_i(\mathbb{E}[R_m] - r_f)$.
- **CML** (total risk, efficient portfolios) $\neq$ **SML** (beta, every asset).

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Discuss (2 minutes)

1. Two stocks, same $\sigma = 25\%$. $\beta_X = 1.4$, $\beta_Y = 0.5$. Which has higher $\mathbb{E}[R]$?

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2. Why can $\beta_i = 0$ still be individually volatile, yet earn only r_f?
3. Why plot individual stocks on the SML but not the CML?

Looking Ahead

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Interpreting Beta & the SML – Lecture 11